

WASP-39b

R-1583

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_160524 · created 2026-08-02T00:09:23+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457532.816 +/- 0.04 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.189 +/- 0.055
Transit depth (Rp/Rs)^2	3.58 +/- 2.06 [%]
Orbital Inclination (inc)	87.3 +/- 2.14
Airmass coefficient 1 (a1)	0.72 +/- 0.42
Airmass coefficient 2 (a2)	0.18 +/- 0.12
Scatter in the residuals of the lightcurve fit is	61.63 %
Best Comparison Star	#1 - [405, 50]
Optimal Aperture	1.64
Optimal Annulus	12.29
Transit Duration (day)	0.095 +/- 0.037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.189 ± 0.055 vs 0.1457 (deviation 0.79σ)
Duration fit vs published	0.095 d vs 0.12 d (deviation 0.68σ)
Mid-time O–C	-7.1 min
Depth SNR	3.4
Residual scatter	61.63 %

Light curve

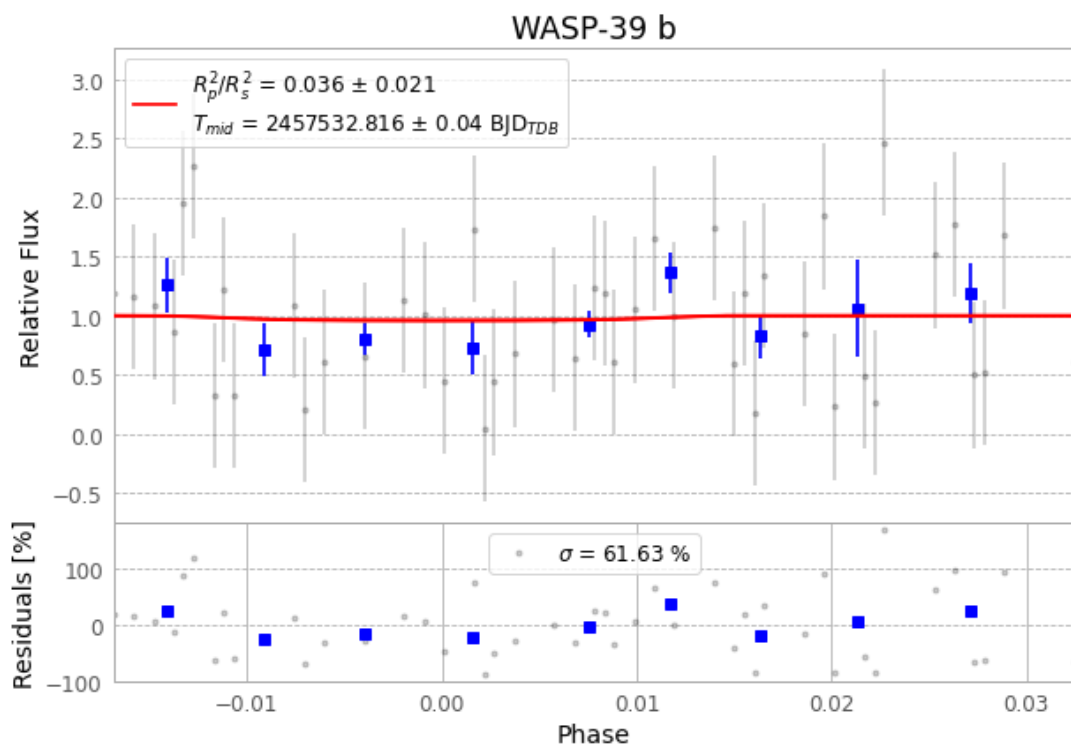


Figure 1 — FinalLightCurve_WASP-39 b_2016-05-23.png (SHA-256 b0d0cac6a33d319c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [130, 389], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 49d5fff3376ffee5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

95 FITS frames (62 MB), WASP-39160524054812.FITS → WASP-39160524103612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-160524.log (25dbd14a2555...), FinalLightCurve_WASP-39 b_2016-05-23.png (b0d0cac6a33d...), FinalLightCurve_WASP-39 b_2016-05-23.csv (4c012cf94365...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 00:09Z.